Total No. of Questions : 10]

SEAT No. :

P974

[Total No. of Pages : 2

[5871]-702

B.E. (Computer Engineering)

ARTIFICIAL INTELLIGENCE AND ROBOTICS

(2015 Pattern) (Semester - I)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Assume suitable data if necessary.***
5) ***Justify your answer with an example wherever necessary.***

Q1) a) Explain constraint satisfaction with examples. **[8]**

b) Give and Explain Hill Climbing algorithm. **[6]**

OR

Q2) a) How planning strategies are classified. Explain in detail. **[8]**

b) Explain Constraint Satisfaction with N-Queens problem. **[6]**

Q3) a) Give Algorithm for Backtracking and Look ahead Strategies of Constraint Satisfaction. **[8]**

b) Explain Knowledge Based Reasoning with details. **[6]**

OR

Q4) a) With examples explain Propositional and First Order Logic. **[8]**

b) Explain Deductive Retrieval with Backward Chaining in details. **[6]**

Q5) a) Explain all Stages in natural language Processing with examples. **[8]**

b) Explain Application of NLP in Machine Translation, Information Retrieval and Big Data Information Retrieval. **[6]**

OR

P.T.O.

Q6) a) Write a short note on Supervised, Unsupervised and Reinforcement learning. **[8]**

b) Explain Feed forward and Feedback ANNs. **[6]**

Q7) a) Explain Sensing and mapping for Point Robot. **[8]**

b) Explain Non Visual Sensors like : Contact Sensors and Inertial Sensors. **[6]**

OR

Q8) a) Draw and Explain Hybrid Control Architecture. **[8]**

b) Write a short note on Human-Robot Interface. **[6]**

Q9) a) Explain in details laser Rangefinders and Biological Sensing. **[8]**

b) Explain Robot Pose Maintenance and Localization. **[6]**

OR

Q10) a) Write a short note on Topological Maps and Geometric Maps. **[8]**

b) Explain in details Intelligent Vehicles. **[6]**

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